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Unit 1 PYQs > | Unit 2 PYQs > | Unit 3 PYQs > | Unit 4 PYQs > | Unit 5 PYQs >

Focus on Unit 2 (Differential Calculus-I) and Unit 4 (Multiple Integration). Together they account for about 63 marks among all questions printed per paper. Unit 3 (Differential Calculus-II) is safest to skim if you're short on time.

Full subject strategy >

TOP UNITS See all >

Unit 2 - Differential Calculus-I Notes ↗ Important questions ↗

Unit 4 - Multiple Integration Important questions ↗

PAPER ID-311443



Subject Code: BAS103

Roll No: | | | | | | | | | | | | | |

**BTECH**
**(SEM I) THEORY EXAMINATION 2023-24**
**ENGINEERING MATHEMATICS-I**

**TIME: 3HRS** **M.MARKS: 70**

**Note: 1.** Attempt all Sections. If require any missing data; then choose suitably.

## SECTION A

**1. Attempt *all* questions in brief.** **2 x 7 = 14**

| Q no. | Question | Marks | CO |
|---|---|---|---|
| a. | Find the product and sum of the eigen values for $A = \begin{bmatrix} 8 & -4 \\ 2 & 2 \end{bmatrix}$. | 2 | 1 |
| b. | Find all symmetry in the curve $y^2(a^2 + x^2) = x^2(a^2 - x^2)$. | 2 | 2 |
| c. | Calculate the error in $R$ if $E = RI$ and possible errors in $E$ and $I$ are 30% and 20% respectively. | 2 | 3 |
| d. | Determine the value of $\Gamma\frac{1}{4}\ \Gamma\frac{3}{4}$. | 2 | 4 |
| e. | Prove that $B(p,q) = B(p+1,q) + B(p,q+1)$ | 2 | 4 |
| f. | Prove that $\vec{A} = (6xy + z^3)\hat{\imath} + (3x^2 - z)\hat{\jmath} + (3xz^2 - y)\hat{k}$ is irrotational. | 2 | 5 |
| g. | Find a unit normal vector to the surface $xy^3z^2 = 4$ at the point $(-1,-1,2)$. | 2 | 5 |

## SECTION B

**2. Attempt any *three* of the following:** **7 x 3 = 21**

| | | | |
|---|---|---|---|
| a. | Solve the system of homogenous equations:<br>$x_1 + x_2 + x_3 + x_4 = 0,\ x_1 + 3x_2 + 2x_3 + 4x_4 = 0,$<br>$2x_1 + x_3 - x_4 = 0$ | 7 | 1 |
| b. | If $u = y^2e^{y/x} + x^2\tan^{-1}\left(\frac{x}{y}\right)$, show that<br>(i) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2u$<br>(ii) $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 2u.$ | 7 | 2 |
| c. | Expand $f(x,y) = e^x\cos y$ about the point $\left(1,\frac{\pi}{4}\right)$ by Taylor's sereies. | 7 | 3 |
| d. | Evaluate the integral $\iint_D (y-x)dxdy$; by changing the variables, D: Region in xy-plane bounded by the lines<br>$y - x = -3, y - x = 1, y + \frac{1}{3}x = \frac{7}{3}, y + \frac{1}{3}x = 5.$ | 7 | 4 |
| e. | Find the directional derivative of $f(x,y,z) = e^{2x}\cos yz$ at $(0,0,0)$ in the direction of the tangent to the curve<br>$x = a\sin\theta, y = a\cos\theta,\quad z = a\theta$ at $\theta = \pi/4$. | 7 | 5 |

## SECTION C

**3. Attempt any *one* part of the following:** **7 x 1 = 7**

| | | | |
|---|---|---|---|
| a. | Determine eigen vectors for the matrix $A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$. | 7 | 1 |
| b. | Determine $A^{-1}, A^{-2}$ and $A^{-3}$ if $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$ using Cayley-Hamilton theorem. | 7 | 1 |



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**4. Attempt any *one* part of the following:** **7 x 1 = 7**

| | | | |
|---|---|---|---|
| a. | If $y = \cos(m \sin^{-1} x)$ then prove that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2 - n^2)y_n = 0$. Also find $(y_n)_0$. | 7 | 2 |
| b. | If $z = f(x,y), x = e^u + e^{-v}, y = e^{-u} - e^v$ then show that $\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x\frac{\partial z}{\partial x} - y\frac{\partial z}{\partial y}$ | 7 | 2 |

**5. Attempt any *one* part of the following:** **7 x 1 = 7**

| | | | |
|---|---|---|---|
| a. | If $u^3 + v^3 = x + y, u^2 + v^2 = x^3 + y^3$, then show $\frac{\partial(u,v)}{\partial(x,y)} = \frac{y^2 - x^2}{2uv(u-v)}$. | 7 | 3 |
| b. | The pressure P at any point $(x,y,z)$ in space is $P = 400\,xyz^2$. Find the highest pressure at the surface of a unit sphere $x^2 + y^2 + z^2 = 1$ using Lagrange's method. | 7 | 3 |

**6. Attempt any *one* part of the following:** **7 x 1 = 7**

| | | | |
|---|---|---|---|
| a. | Find the volume of the solid bounded by the coordinate planes and the surface $\left(\frac{x}{a}\right)^{1/2} + \left(\frac{y}{b}\right)^{1/2} + \left(\frac{z}{c}\right)^{1/2} = 1$. | 7 | 4 |
| b. | Prove that $B(m,n) = \frac{\Gamma m \Gamma n}{\Gamma m+n}$. | 7 | 4 |

**7. Attempt any *one* part of the following:** **7 x 1 = 7**

| | | | |
|---|---|---|---|
| a. | Applying Gauss Divergence theorem, evaluate $\iint_S [e^x dydz - ye^x dzdx + 3zdxdy]$, where S is the surface of the cylinder $x^2 + y^2 = c^2, 0 \le z \le h$. | 7 | 5 |
| b. | Prove that $\nabla^2 r^n = n(n+1)r^{n-2}$, where $\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$ and hence show that $\nabla^2\left(\frac{1}{r}\right) = 0$. | 7 | 5 |

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AKTU - M1 - ALL UNITS - FROM 8 PAST PAPERS

# Top 3 AKTU questions - all units

Engineering Mathematics-i. Done with the paper? These come back the most across the whole subject.

1 Change the order of integration and hence evaluate $\int_0^a\int_{\sqrt{ax}}^a\frac{y^2}{\sqrt{y^4-a^2x^2}}\,dy\,dx$.

- Unit 4 - 7 marks - asked 7 times - last 2026

2 If $z = x^4y^2 \sin^{-1}(\frac{x}{y}) + \log x - \log y$, show that $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 6x^4y^2 \sin^{-1}(\frac{x}{y})$.

- Unit 2 - 7 marks - asked 7 times - last 2026

3 If $y = e^{\tan^{-1}x}$, prove that $(1 + x^2)y_{n+2} + [(2n+2)x - 1]y_{n+1} + n(n+1)y_n = 0$.

- Unit 2 - 10 marks - asked 8 times - last 2026

**Practise all important questions >**

Tick 'I can answer this' and track your prep.

## EVERY UNIT - NOTES, IMP QS, PYQS, STRATEGY

Tap a link to open resources for that unit.

| | | | | |
|---|---|---|---|---|
| **Unit 1 - Matrices** | Notes ↗ | IMP Qs ↗ | PYQs ↗ | Strategy ↗ |
| **Unit 2 - Differential Calculus-I** | Notes ↗ | IMP Qs ↗ | PYQs ↗ | Strategy ↗ |
| **Unit 3 - Differential Calculus-II** | Notes ↗ | IMP Qs ↗ | PYQs ↗ | Strategy ↗ |
| **Unit 4 - Multiple Integration** | | IMP Qs ↗ | PYQs ↗ | Strategy ↗ |
| **Unit 5 - Vector Calculus** | | IMP Qs ↗ | PYQs ↗ | Strategy ↗ |

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AKTU - M1

# Paper Done.

## See What Repeats InEngineering Mathematics-i

**187**
QUESTIONS ANALYSED

We analysed 8 AKTU papers - these questions keep coming back.

- **8 AKTU papers analysed | 187 questions | 2019-2026**

### Must-do topics & questions
Questions that repeat in AKTU papers, ranked unit-wise. **Open >**

### Predicted paper
A likely paper built from past AKTU papers. Practise it before the exam. **Open >**

### Past papers (PYQ)
Filter by year, unit and topic. See which questions repeat. **Open >**

### Subject strategy
Unit weightage, safest units and what to study first. **Open >**

### Ask Ark - AI made for AKTU
Reads real AKTU questions and AcademicArk notes. Ask any doubt. **Open >**

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